Improving Association Rule Mining using Clustering-Based Item Similarity

Itamar Berger

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1 Abstract

Association rule mining (ARM) is widely used for discovering relationships between items in large datasets, but traditional methods like the Apriori and the FPGrowth algorithms face challenges related to high computational cost and relevance of generated rules [5].

In this paper, we propose an approach to improve ARM by integrating clustering-based item similarity. This method clusters transactions with similar patterns, reducing the search space and focusing on generating more meaningful rules. The clusters are formed using KMeans, which helps identify relationships between items that might be overlooked in conventional frequency-based approaches, as demonstrated in [6]. We evaluate our method using 4 well-known datasets and compare it with traditional ARM techniques and specifically the Apriori algorithm.

Experimental results show that our clustering-based approach significantly enhances the quality of the association rules, providing more relevant insights while maintaining computational efficiency. Our findings suggest that clustering-based item similarity is an effective enhancement for traditional ARM methods, offering a scalable solution for discovering relevant and actionable item relationships in large datasets.

2 Problem Description

The primary focus of this work is improving the Association Rule Mining (ARM) process, specifically targeting the improvement of lift values in traditional rule generation methods. In conventional ARM algorithms, such as Apriori, rules are mined based on item frequency, which can often lead to the generation of rules with lower lift values, indicating weak associations between items.

A key limitation of traditional methods is that they do not account for the similarity between transactions, which can significantly impact the quality of the generated rules. Treating transactions independently often leads to the generation of trivial rules with sub-optimal lift values. Furthermore, traditional approaches often overlook rare but potentially valuable rules, as they may not consider lower support thresholds due to computational constraints or because doing so leads to overall low lift values.

Our approach addresses this by clustering transactions based on their similarity. By grouping similar transactions together, we aim to focus rule generation on clusters with stronger associations, leading to improved lift values. By exploiting patterns in similar transactional behaviors, this method enhances the quality and relevance of the resulting rules, ultimately increasing lift and the overall strength of associations, while maintaining computational efficiency.

3 Solution Overview

Our approach begins by clustering the transactions using the KMeans algorithm [7], where the number of clusters k is determined through the elbow method explained in here [8]. Since the transactional data is represented as binary vectors, KMeans is particularly suitable for this task. This is because the KMeans algorithm uses Euclidean metric which is:

$$d(A,B) = \sqrt{\sum_{i=1}^{n}(a_i - b_i)^2} \tag{1}$$

in our case all of the transactions are represented as binary vectors:

For each i, $\mathrm{A}_i = \begin{cases} 1 & \text{if item i is present} \\ 0 & \text{o.w} \end{cases}$

Namely, we can conclude that our Euclidean distance represents the square root of the number of items that differ between the two transactions:

$$d(A,B) = \sqrt{\sum_{i=1}^{n}(a_i - b_i)^2} = \sqrt{\text{Hamming Distance}(A,B)} \tag{2}$$

Thus, transactions are grouped effectively based on similarity, where transactions with common items will be grouped together.

In order to enable effective clustering and avoid the curse of dimensionality, we employ a heuristic approach for dimension reduction. We represent each transaction by selecting the top k% most popular items across the entire dataset. Our empirical analysis shows that selecting k between 1% and 20% achieves a good balance of accuracy and efficiency. This method effectively captures the core behavior or overall patterns of the transaction, while significantly reducing its dimensionality.

Once the transactions are clustered, we apply the Apriori algorithm to each cluster separately, using higher support thresholds to better suit the smaller clusters datasets and avoid noisy unreliable rules. This allows for more focused and relevant rule generation within each cluster.

To improve clustering efficiency and mitigate the curse of dimensionality, we apply a heuristic dimensionality reduction technique. Instead of using all items, we retain only the top $k\%$ most frequently occurring items across the dataset. Empirical analysis shows that selecting k between 1% and 20% achieves an optimal balance between accuracy and efficiency. This reduction strategy enhances clustering performance by eliminating noise, improving computational scalability and clusters distribution.

We use the FPGrowth algorithm instead of Apriori to improve computational efficiency, as it avoids costly candidate generation while still identifying the same frequent itemsets and association rules. Given their equivalence in this context [9], we use the terms interchangeably .

By using a hybrid approach, combining KMeans clustering with FPGrowth/Apriori, we create a process that not only ensures computational feasibility but also enhances the relevance and quality of the generated association rules.

4 Experimental Evaluation

To assess the effectiveness of our solution, we compared it against the baseline approach using three key metrics commonly used in association rule mining: **Support**, **Confidence**, and **Lift**. These metrics provide insights into the strength and relevance of the generated association rules. Our objective was to demonstrate that our proposed approach enhances rule quality compared to the baseline.

4.1 Datasets

We conducted experiments on four publicly available datasets, transforming each into a transactional format suitable for association rule mining:

- **MovieLens 20M**: Contains 20 million movie ratings from 138,493 users across 27,278 movies. This dataset was obtained from Kaggle (available at https://www.kaggle.com/datasets/grouplens/movielens-20m-dataset). We converted it into a transactional dataset by representing each user as a transaction containing the movies they rated.

- **Online Retail Dataset**: Consists of purchase records from 4,000 customers, with over 25,000 transactions. Also obtained from Kaggle (available at https://www.kaggle.com/datasets/vijayuv/onlineretail), this dataset was transformed into a transactional format by aggregating each customer's entire shopping history into a single transaction.

- **Last.FM Dataset**: Contains 92,800 artist listening records from 1,892 users. This dataset was obtained from the GroupLens website (https://grouplens.org/datasets/hetrec-2011/). Each user's listening history was treated as a transaction.

- **Netflix Prize Data**: Includes movie-watching histories of 480,189 users. This dataset was sourced from Kaggle (https://www.kaggle.com/datasets/netflix-inc/netflix-prize-data), and transactions were created by grouping all movies watched by a user.

4.2 Evaluation Methods

For each dataset, we applied the following methods:

- **Baseline Approach**: The traditional Apriori algorithm, which extracts rules from the entire dataset without considering transaction similarity.
- **Proposed Approach**: Our method, which clusters transactions using KMeans and applies the Apriori algorithm separately within each cluster, using higher support thresholds tailored to the smaller clustered datasets. The final rule set is obtained by merging distinct rules across clusters.

4.3 Results

4.3.1 MovieLens 20M Dataset

Table 1: Baseline Approach (min support = 0.18)
Number of rules: 3176

Metric	**Avg**	**Min**	**Max**
Support	0.200921	0.18	0.3542
Confidence	0.625521	0.3605	0.9566
Lift	**1.8707**	1.1664	3.7134

Table 2: Our Solution (in-cluster min support = 0.5)
Number of rules: 9484

Metric	**Avg**	**Min**	**Max**
Support	0.1965	0.0354	0.34865
Confidence	0.6443	0.1025	0.981992
Lift	**2.3092**	1.2140	7.6053

The comparison shows that our approach increases the Lift from 1.87 to 2.31, indicating stronger relationships between items. Although the Support and Confidence show slight variations, the increase in Lift is substantial, suggesting the uncovering of more significant associations.

4.3.2 Online Retail Dataset

Our approach shows a noticeable increase in Lift (from 5.68 to 8.22) and a slight increase in Confidence (from 0.571 to 0.572). The minimal change in Confidence

Table 3: Baseline Approach (min support = 0.05)
Number of rules: 240

Metric	Avg	Min	Max
Support	0.0580349	0.05	0.0762791
Confidence	0.570955	0.252619	0.965957
Lift	**5.67581**	1.48258	11.8352

Table 4: Our Solution (in-cluster min support = 0.4)
Number of rules: 1852

Metric	Avg	Min	Max
Support	0.0439065	0.0253488	0.0755814
Confidence	0.572033	0.129505	1.0000000
Lift	**8.22112**	2.12599	29.2759

suggests that the generated rules maintain a similar level of certainty as the baseline, but the substantial rise in Lift indicates that our method uncovers stronger, more significant relationships. We suggest that the focus on clustering by transaction similarity allows for the identification of meaningful patterns across varying levels of transaction frequency, offering valuable insights into customer behavior.

4.3.3 Last.FM Dataset

Table 5: Baseline Approach (min support = 0.099)
Number of rules: 4458

Metric	Avg	Min	Max
Support	0.1159801	0.1002191	0.2387733
Confidence	0.6656595	0.2995090	0.9947368
Lift	**3.6084335**	1.6627986	5.0973262

Table 6: Our Solution (in-cluster min support = 0.4)
Number of rules: 2242

Metric	Avg	Min	Max
Support	0.0238546	0.0093100	0.2168675
Confidence	0.3884072	0.0278232	1.0000000
Lift	**5.5031059**	0.6973268	17.1993721

Our approach results in an increase in Lift (from 3.61 to 5.50) and a decrease in Confidence (from 0.665 to 0.388), indicating a tradeoff between the strength and certainty of the associations. While the lower Confidence suggests that some rules are less certain, the increase in Lift demonstrates that these rules capture stronger and more meaningful relationships between items. This tradeoff is a common outcome when focusing on rare but impactful rules.

4.3.4 Netflix Prize Dataset

Table 7: Baseline Approach (min support = 0.09)
Number of rules: 112

Metric	**Avg**	**Min**	**Max**
Support	0.0330308	0.0108033	0.3456590
Confidence	0.7132389	0.0434620	1.0000000
Lift	**21.4071420**	2.8930248	92.5643777

Table 8: Our Solution (in-cluster min support = 0.02)
Number of rules: 616

Metric	**Avg**	**Min**	**Max**
Support	0.0060682	0.0010896	0.3456590
Confidence	0.3343641	0.0031522	1.0000000
Lift	**37.0355164**	0.1391973	917.7659574

Our approach results in a dramatic increase in Lift (from 21.41 to 37.04) and a decrease in Confidence (from 0.713 to 0.334). This further emphasizes the tradeoff between the strength and certainty of relationships. The significant increase in Lift demonstrates that the new rules capture far stronger associations than the baseline. This trend also highlights that focusing on less frequent but impactful patterns can result in highly meaningful, though less certain, rules.

4.3.5 Summary

In summary, our approach clearly outperforms the baseline in generating association rules with higher lift. Moreover, all results were obtained in less time than the baseline, demonstrating that our solution is at least as scalable as the standard Apriori algorithm—if not more—given that it produces significantly more rules in 3 out of the 4 datasets.

That being said, achieving higher lift comes at the cost of lower overall support across all four datasets. However, this does not undermine our approach, as we ensured that the minimum support threshold was maintained within each cluster, guaranteeing that every generated rule has strong support within its similarity group.

Regarding confidence, there is no definitive trend: it remains stable in the first two datasets but declines in the last two.

5 Related Work

Dharshinni [1] used KMeans clustering based on attributes like gender and age, demonstrating that applying the Apriori algorithm within each cluster improved the quality of association rules by providing higher confidence and better time efficiency. M. Harahap et al [6], applies the k-means algorithm to clustering the 10 most dominant diseases and look for patterns. The concept of combining clustering with association rule mining forms the foundation of our approach. However, while previous studies primarily focused on comparing the association rules generated within each cluster to those produced using the baseline method for the same cluster, our approach extends this analysis by evaluating and comparing the generated rule sets not only at the cluster level but also as a unified whole.

Building on this, our method extends clustering by incorporating user behavior as the clustering criterion, similar to user-based collaborative filtering (CF) methods [2]. Unlike Dharshinni's demographic-based clustering, we focus on grouping users based solely on their transaction or purchase history, providing a more effective solution even without relying on prior user knowledge.

Zhang [4] introduced a covering-based collaborative filtering approach that builds user profiles using a subset of highly rated items, demonstrating improvements in recommendation accuracy. While their work was limited to the MovieLens dataset, we extend this concept to multiple datasets with an emphasis on dimensionality reduction.

Unlike Zhang's approach, which selects items based on user ratings, we globally identify the most significant items across all users and retain a subset to represent each user. This reduces dimensionality while preserving meaningful similarities. Our analysis shows that selecting the top 1% to 20% of items optimizes clustering, balancing accuracy and efficiency.

6 Conclusion

In this paper, we presented a novel approach to improving Association Rule Mining (ARM) by incorporating clustering-based item similarity. By clustering transactions based on their similarity, we effectively reduced the search space, allowing the Apriori algorithm to generate more meaningful and relevant association rules. Our experiments demonstrated that this approach significantly enhances the quality of generated rules, particularly in terms of Lift, indicating stronger relationships between items.

The results from multiple datasets showed that our method not only improved rule quality but also maintained computational efficiency, making it a scalable solution for large datasets. Although the trade-off between Lift and Confidence was observed, this is a natural outcome when focusing on less frequent but impactful associations. Overall, our clustering-based approach offers a powerful enhancement to traditional ARM methods, providing an effective alternative to discover valuable item relationships in complex datasets.

References

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